DSA210 Final Report: Analyzing the Long-term Impact of COVID-19 on Academic Performance

Motivation

The COVID-19 pandemic caused unprecedented disruptions to global education systems. The motivation behind this project is to quantitatively measure the true extent of the pandemic's "learning loss" and structural disruption. Rather than relying on simple pre- and post-pandemic averages, this project aims to model the trajectory of academic performance across three distinct phases: Pre-COVID (2017–2019), Lockdown/Transition (2020–2022), and Recovery (2023–2025). By applying both classical statistical methods and machine learning techniques, the project seeks to answer two core questions:

1. Did the pandemic cause a measurable, statistically significant change in student academic outcomes?
2. Which factors (economic disadvantage, demographic composition, policy stringency) are most strongly associated with learning loss and recovery?

Understanding these patterns can help policymakers allocate resources more effectively and design targeted interventions for schools and districts that are struggling to recover.

Data Sources

Primary Dataset

The primary dataset was obtained from the Stanford Education Data Archive (SEDA), which aggregates district-level academic performance and demographic information for U.S. public schools.

Source: https://cepa.stanford.edu/seda2/data-download

File used: seda2023_cov_state_annual.csv (provided in the project repository)

Collection method: The CSV file was downloaded directly and loaded into a Jupyter notebook using pandas.read_csv().

Time period extracted: 2016–2022 (2016 and 2017 included for trend analysis; core comparisons focused on 2019 vs. 2022)

Data characteristics:

255 rows (state-year observations)

11 columns, including:

sedafips – state FIPS code

year – academic year

totenrl – total enrollment

perfrl – percentage of students eligible for free/reduced-price lunch

perblk, perhsp, perasn, pernam, peroth, perwht – racial/ethnic percentages

Secondary Dataset

For machine learning analysis, a separate dataset was used, focusing on career and technical education (CTE) programs in Pennsylvania.

Source: Provided as ffx8-fm96_version_20.csv

Time period: 2020–2023 (school years 2020-2021 through 2023-2024)

Data characteristics after cleaning:

388 rows

8 numeric feature columns (e.g., _5s4_technical_skill, _2s1_academic_proficiency, _1s1_four_year_graduation)

Target variable: _2s2_academic_proficiency (a measure of academic proficiency)

Cleaning steps:

Dropped non-numeric or irrelevant columns (_2s3_academic_proficiency, fiscal_agent_name, fiscal_agent_aun)

Removed rows with missing values

Mapped school years to integer values for chronological splitting

Enrichment (Planned but not implemented)

The project proposal included enrichment with the Oxford COVID-19 Government Response Tracker (OxCGRT) stringency index to capture lockdown intensity. While this enrichment was not completed within the submitted notebooks, the planned approach was to merge the stringency index by state and year, allowing analysis of how policy strictness moderates academic outcomes.

Data Analysis

The analysis was conducted in two major phases: Exploratory Data Analysis & Hypothesis Testing (using the SEDA dataset) and Machine Learning Modeling (using the Pennsylvania dataset).

Exploratory Data Analysis

Enrollment Trends

A line plot of mean total enrollment from 2016 to 2022 showed a steady increase from 2016 to 2019, followed by a sharp decline in 2022. A boxplot confirmed that the distribution of enrollment across states shifted downward in 2022 compared to previous years.

Correlation Heatmap

A correlation matrix of all numeric variables revealed expected relationships:

perwht (percent white) was strongly negatively correlated with perhsp (percent Hispanic) and **perfrl** (percent free/reduced lunch).

perfrl was positively correlated with perblk and negatively correlated with perwht.

totenrl (total enrollment) showed moderate positive correlation with enrl38 (grades 3-8 enrollment), as expected.

Time Series for a Single State (California, FIP 6)

A line plot for California showed enrollment stable around 6.2 million from 2016–2018, with a minor dip in 2019 and a sharp drop to approximately 5.87 million in 2022, supporting the premise that the pandemic disrupted public school enrollment.

Hypothesis Testing

Hypothesis 1 (H1): Change in Economic Disadvantage

Null hypothesis (H_0): No significant difference in perfrl (free/reduced-price lunch percentage) between 2019 and 2022.

Test used: Paired t-test and Wilcoxon signed-rank test.

Results:

Mean perfrl 2019: 0.4885

Mean perfrl 2022: 0.4596

Paired t-test: $t = 3.6874$, $p = 5.58\times10^{-4}$

Wilcoxon test: $p = 7.43\times10^{-8}$

Interpretation: The null hypothesis was rejected. The percentage of students eligible for free/reduced-price lunch decreased significantly from 2019 to 2022. While this contradicts a simple "worsening economy" narrative, it may reflect federal universal free lunch waivers during COVID-19, which reduced the need for families to complete paperwork to be counted in this metric.

Hypothesis 3 (H3): Change in Racial/Ethnic Composition

Null hypothesis (H_0): No significant difference in demographic percentages between 2019 and 2022.

Tests used: Paired t-tests for each demographic variable.

Results (summarized):

Demographic	Mean 2019	Mean 2022	Change	p-Value
perwht(White)	**0.5614**	**0.5437**	**-0.0177**	**3.10×10^{-19}**
perblk(Black)	**0.1429**	**0.1408**	**-0.0021**	**9.14×10^{-3}**
perhsp(Hispanic)	**0.1815**	**0.1946**	**+0.0131**	**9.00×10^{-19}**
perasn(Asian)	**0.0493**	**0.0492**	**-0.0001**	**0.905 (not significant)**
pernam(Native)	**0.0211**	**0.0201**	**-0.0010**	**0.0117**
peroth(other)	**0.0439**	**0.0515**	**+0.0076**	**1.18×10^{-16}**

Interpretation: Significant decreases were observed in the proportion of white and Black students, while Hispanic and "Other" categories increased significantly. The Asian student proportion remained stable. These shifts suggest that the pandemic may have influenced geographic mobility, school choice, or differential enrollment patterns across racial/ethnic groups.

Machine Learning Analysis

The goal was to predict academic proficiency (_2s2_academic_proficiency) using other performance metrics (e.g., technical skills, reading proficiency, graduation rates) as features. A chronological split was used to simulate a real-world forecasting scenario:

Training set: School years 2020-2021 and 2021-2022 (194 rows)

Validation set: School year 2022-2023 (97 rows)

Test set: School year 2023-2024 (97 rows)

Models and Hyper parameter Tuning

1. Linear Regression (baseline)
2. Ridge Regression (tuned with alpha from 0.001 to 1000)
3. K-Nearest Neighbors (KNN) Regressor (tuned for n_neighbors and weights)
4. Random Forest Regressor (tuned for n_estimators, max_depth, min_samples_leaf)

Hyperparameter tuning was performed using GridSearchCV with a PredefinedSplit that treated the training and validation sets separately, respecting chronological order.

Model Performance

Model	Train RMSE	Train R^2	Val RMSE	Val R^2	Test RMSE	Test R^2
Linear Regression	**0.200**	**0.235**	**0.239**	**-0.732**	**0.181**	**-0.089**
Ridge Regression	**0.220**	**0.074**	**0.232**	**-0.636**	**0.165**	**0.089**
KNN Regressor	**0.189**	**0.319**	**0.213**	**-0.379**	**0.132**	**0.418**
Random Forest	**0.124**	**0.707**	**0.226**	**-0.554**	**0.129**	**0.445**

Interpretation

- All models performed poorly on the validation set (2022-2023), with negative R^2 values. This indicates that the 2022-2023 school year was an anomalous "transition year" where historical relationships between features and the target broke down.
- On the test set (2023-2024), both KNN and Random Forest recovered strongly, achieving R^2 values of 0.418 and 0.445, respectively. Linear models remained poor, suggesting that the underlying relationships are non-linear.
- Random Forest achieved the lowest test RMSE (0.129) and highest test R^2 (0.445), making it the best overall model.

Findings

Statistical Findings (SEDA Dataset)

1. **Enrollment dropped significantly** from 2019 to 2022 (paired t-test $p < 0.001$), supporting the hypothesis that the pandemic disrupted student participation in public education.

2. **Economic disadvantage, as measured by free/reduced-price lunch eligibility, decreased significantly** from 2019 to 2022. This counterintuitive result is likely explained by federal universal free lunch waivers that removed the need for families to apply, rather than an actual improvement in economic conditions.

3. Racial/ethnic composition shifted significantly:
    - White student percentage decreased ($p < 10^{-18}$)
    - Hispanic student percentage increased ($p < 10^{-18}$)
    - “Other” category increased ($p < 10^{-16}$)
    - Black student percentage decreased modestly ($p = 0.009$)
    - Asian student percentage did not change significantly ($p = 0.905$)
4. **Correlation heatmaps** (2019 vs. 2022) showed that the overall structure of relationships among demographic variables remained similar across the two time points.

Machine Learning Findings (Pennsylvania Dataset)

1. The 2022-2023 school year was a chaotic “transition year.” All models produced negative R^2 values on the validation set, meaning they performed worse than simply predicting the mean. This suggests that the relationships between features (e.g., technical skills, reading proficiency) and academic proficiency were fundamentally disrupted as schools returned to in-person learning.

2. By 2023-2024, academic outcomes restabilized. Non-linear models (KNN and Random Forest) achieved R^2 values of approximately 0.42–0.45 on the test set, indicating that predictable patterns had re-emerged.

3. Linear models failed to capture the underlying structure. Linear Regression and Ridge Regression performed poorly on both validation and test sets, confirming that the drivers of academic proficiency are not simply additive or linear.

4. Random Forest was the best overall model, with the lowest test RMSE (0.129) and highest test R^2 (0.445).

Limitations and Future Work

Limitations

1. **Incomplete enrichment:** The planned enrichment with the Oxford COVID-19 Government Response Tracker (OxCGRT) stringency index was not implemented in the submitted notebooks. Without this, the analysis cannot directly model how varying lockdown intensities affected learning loss.

2. **Different datasets for different analyses:** The hypothesis testing was performed on the SEDA national dataset, while the machine learning analysis used a separate Pennsylvania-specific dataset. This makes it difficult to directly compare findings or generalize ML results nationally.

3. **Limited time range in SEDA data:** The available SEDA data only extended to 2022, preventing analysis of the 2023–2025 “recovery period” as originally proposed.

4. **Small feature set for ML:** The Pennsylvania dataset contained only six numeric features, which may not capture all relevant factors influencing academic proficiency (e.g., internet access at home, teacher-to-student ratios, local economic conditions).

5. **Potential overfitting:** While cross-validation was used, the Random Forest model showed a large gap between train R^2 (0.707) and test R^2 (0.445), suggesting some overfitting to the training period.

6. **Lack of causal inference:** The statistical tests establish associations, not causation. Other unmeasured confounders (e.g., changes in testing policies, student mobility) may explain some of the observed changes.

Future Work

1. **Implement the OxCGRT enrichment** to directly measure the impact of lockdown stringency on academic outcomes across states.

2. **Extend the SEDA time series to include 2023–2025** data once available, enabling a true pre-/during-/post-COVID comparison.

3. **Apply the machine learning pipeline** to the national SEDA dataset to assess whether the patterns observed in Pennsylvania hold nationwide.

4. **Incorporate additional features such as:**
    - Broadband internet access at the district level
    - Teacher turnover rates
    - School funding per pupil
    - Proportion of students learning remotely during 2020-2022

5. **Use clustering (e.g., K-Means)** as originally proposed to categorize districts into "Resilient," "Recovering," or "Struggling" groups based on their post-COVID recovery trajectories.

6. **Perform an Interrupted Time Series (ITS)** analysis to formally estimate the causal impact of the pandemic by forecasting a counterfactual "no-pandemic" trend using pre-2020 data.

7. **Explore feature importance in the Random Forest model** to identify which factors (e.g., reading proficiency, technical skills, graduation rates) are most predictive of math proficiency in the post-pandemic era.

References

Reardon, S. F., et al. (2023). Stanford Education Data Archive (Version SEDA2023). https://doi.org/10.25740/xt779fj2637

Pennsylvania Department of Education. CTE Performance Data (ffx8-fm96_version_20.csv).